PROJECT ABSTRACT

PowerPulse

An Adaptive Low-Power RISC-V SoC with Hardware Activity Profiling and Autonomous Peripheral Wake-Up

1. Problem Statement

Modern embedded System-on-Chip (SoC) designs integrate several peripherals for communication, timing and control, yet these peripherals are not continuously active in most operating scenarios. Keeping every peripheral fully enabled during long idle intervals increases unnecessary switching activity and provides no mechanism for hardware-assisted, power-aware operation. A purely software-controlled approach depends on repeated processor polling, which increases processor activity and delays the response to asynchronous events.

There is therefore a need for an RTL-based SoC architecture that can observe peripheral activity, detect programmable idle periods, reduce the activity of selected peripherals, and restore normal operation automatically when a valid event occurs — while preserving processor access, maintaining software visibility, and remaining fully demonstrable through simulation and verification.

This project proposes PowerPulse, an adaptive low-power-aware RISC-V System-on-Chip built around the faculty-provided RISC-V processor core. The design retains all mandatory components (RISC-V Processor, Instruction Memory, Data Memory, Bus Interconnect, UART, Timer and GPIO) and integrates three custom IP blocks that together provide autonomous, hardware-driven activity management. The project focuses on RTL-level activity management using controlled peripheral-enable / clock-enable signals, and does not claim technology-dependent physical power gating.

2. Proposed SoC Architecture

The architecture is centred on the faculty-provided RISC-V processor, which executes the embedded application and accesses memories and peripherals through a shared memory-mapped Bus Interconnect. Instruction Memory stores the program image; Data Memory stores runtime variables and application data. Among the mandatory peripherals, UART provides serial communication and diagnostic output, the Timer provides timing and programmable event generation, and GPIO provides configurable digital I/O with external event detection.

The three additional IP blocks provide the adaptive activity-management functionality of the SoC. The Hardware Activity Profiler (HAP) monitors selected peripheral activity and maintains programmable inactivity counters, generating an idle indication once a configured threshold is reached. This indication is consumed by the Peripheral Power Management Controller (PPMC), which applies the configured policy and transitions selected peripherals between ACTIVE, IDLE and LOW-ACTIVITY states via controlled enable signals. The Autonomous Wake-Up Event Controller (AWEC) independently monitors enabled wake-up sources — UART receive activity, GPIO edge events and Timer compare events — and, on a valid event, captures the source and issues a wake-up request that causes the PPMC to restore the affected peripheral to the ACTIVE state.

The complete functional sequence is therefore: Peripheral Activity Monitoring → Idle Detection → Transition to LOW-ACTIVITY → Wake-Up Event Detection → Automatic Peripheral Restoration → Software Status Handling. All blocks are integrated as independent memory-mapped slaves, which reduces dependence on continuous software polling and demonstrates coordinated hardware/software behaviour within a complete processor-based RTL SoC.

3. Selected Additional IPs

3.1 Hardware Activity Profiler (HAP)

A custom monitoring IP that detects activity on selected peripherals and measures inactivity intervals.

- Monitors selected peripheral activity signals and maintains inactivity counters
- Provides a programmable idle threshold; resets counters when activity is detected
- Generates an idle indication once inactivity exceeds the configured threshold
- Exposes memory-mapped configuration and status registers

3.2 Peripheral Power Management Controller (PPMC)

The central state-management IP of the architecture.

- Receives idle indications from the HAP and wake-up requests from the AWEC
- Controls transitions between ACTIVE, IDLE and LOW-ACTIVITY states
- Generates peripheral-enable / clock-enable signals at RTL level
- Provides software control over policy and visibility into current state

3.3 Autonomous Wake-Up Event Controller (AWEC)

A custom event-management IP that monitors multiple enabled wake-up sources (UART receive activity, GPIO edge events, Timer compare events).

- Supports wake-up source enable masking and event detection / capture
- Identifies the wake-source and generates a wake-up request
- Provides event acknowledgement and clearing, avoiding continuous software polling